Auteur: Xander De Blauwer
Studierichting: Informatica
Oefeningenreeks 4A nr. 15.7

$$\int x^3 e^{5x} dx$$

$$\begin{cases} u_1 = x^3 \\ dv_1 = e^{5x} dx \end{cases} \Rightarrow \begin{cases} du_1 = 3x^2 dx \\ v_1 = \dfrac{e^{5x}}{5} \end{cases}$$

$$= \frac{x^3 \cdot e^{5x}}{5} - \int \frac{3}{5} x^2 e^{5x} dx$$

$$\begin{cases} u_2 = \dfrac{3}{5} x^2 \\ dv_2 = e^{5x} dx \end{cases} \Rightarrow \begin{cases} du_2 = \dfrac{6}{5} x dx \\ v_2 = \dfrac{e^{5x}}{5} \end{cases}$$

$$= \frac{x^3 \cdot e^{5x}}{5} - \frac{3x^2 e^5 x}{25} + \int \frac{6}{25} x e^{5x} dx$$

$$\begin{cases} u_3 = \dfrac{6}{25} x \\ dv_3 = e^{5x} dx \end{cases} \Rightarrow \begin{cases} du_3 = \dfrac{6}{25} dx \\ v_3 = \dfrac{e^{5x}}{5} \end{cases}$$

$$= \frac{x^3 e^{5x}}{5} - \frac{3x^2 e^5 x}{25} + \frac{6x e^{5x}}{125} - \frac{6}{625} e^{5x} + c$$

References